Rochelle Pun

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Education

Imperial College London London, UK

MSc Geo-Energy with Machine Learning and Data Science Oct 2024 – Sept 2025

Grade: Merit

- Thesis (completed during TotalEnergies placement): Machine learning approaches to acoustic impedance prediction for exploration-stage reservoir characterisation.
- Awarded the TotalEnergies sponsorship for excellence in geophysical research.

BSc Geophysics Oct 2021 – June 2024

Grade: Upper Second Class Honours (2:1)

- Awarded the Earth Science & Engineering Student Contribution Prize for outstanding contributions to the college and department.

Island School Hong Kong

IB International Baccalaureate Diploma Programme Aug 2019 – June 2021

Scored 44/45 (top 2% worldwide); Higher Levels in Physics, Maths and Geography (all 7s).

Relevant Experience

Data Scientist (MSc Placement) Paris, France

TotalEnergies June 2025 – Nov 2025

- Developed deep learning models (CNNs & RNNs) in PyTorch to predict physical properties from multi-channel time series; replacing key stages of conventional inversion saving over £300k.
- Investigated physics-informed machine learning (PINNs & hybrid physics-ML models); working with geoscientists to inform exploration decisions across eight wells.
- Built a physics-based seismic simulation to generate synthetic training data, addressing data scarcity; integrated into TotalEnergies' codebase adopted by three data science teams.
- Productionised research prototypes into a Python library and CLI for data processing and inference; cutting study turnaround by two weeks across exploration and production teams.

Research Intern Leeds, UK

University of Leeds July 2024 – Sept 2024

- Processed and analysed satellite time series data using HPC (Python/Bash), to investigate crustal deformation; presented findings at the Institute of Geophysics and Tectonics (University of Leeds), and 2025 COMET Student Meeting (University of Bristol).
- Applied statistical modelling (Bayesian inversion via MCMC) to quantify uncertainty; produced the first modern slip-rate estimate of the Herat fault.
- Built an interactive geospatial visualiser; adapted into the open-source LiCSBAS InSAR package, now used in the COMET InSAR training workshop with over 300 annual attendees.

Research Assistant London, UK

Imperial College London June 2024

- Analysed fluid migration pathways using 3D geospatial data to evaluate carbon storage containment risk.

Research Assistant London, UK
Imperial College London, with SAND Geophysics and Geoteric Aug 2023 – Sept 2023

- Interpreted 3D geospatial data to assess foundation hazards for offshore wind-farm sites; contributed to findings presented at three conferences.
- Led weekly presentations to industry leaders SAND Geophysics and Geoteric; translating stakeholder feedback into research priorities and informing a successful PhD proposal to extend the work.

Projects

BSc Thesis
Constraining crustal deformation of the Eastern Pacific Oct 2023 – Dec 2023

- Developed Python tools to process earthquake source mechanisms; subsequently extended during a research internship at the University of Leeds.
- Investigated clustering (k-means, k-medoids, DBSCAN) and KNN classification to assess spatiotemporal distribution of earthquakes.

Conferences & Invited Talks

- Presented at the COMET Student Meeting, University of Bristol. 2025
- Presented at the Institute of Geophysics and Tectonics, University of Leeds. 2024
- Attended the COMET Annual Meeting, University of Huddersfield. 2024
- Attended the GESGB Geophysics Symposium, with findings presented by Dr Rebecca Bell. 2023

Technical Skills

Programming Languages:	Python, Bash, C++
Python Stack:	NumPy, pandas, SciPy, Matplotlib, scikit-learn, PyTorch, TensorFlow, seaborn, Plotly, xarray, netCDF, Shapely, Rasterio, ObsPy
Software Engineering:	Git/GitHub, Linux, packaging, dependency & environment management, code review, debugging, technical documentation

Leadership & Activities

Imperial J-Pop Band Treasurer & Manager
Imperial College Student Union Oct 2022 – May 2025

- Managed a 30-member student band, directing songs from planning to performance at termly concerts with 60-100 ticketed attendees.
- Handled event and administrative operations, eliminating an £800 society deficit to achieve financial break-even.

Student Union Representative
Recreational Clubs Committee, Imperial College Student Union Oct 2022 – June 2023

- Represented 14 student societies, advocating for their interests and rights in discussions with the student union and college leadership.
- Organised networking events and meetings, providing administrative and financial guidance to support society operations.

References available upon request